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A coupled ordinary differential equation lattice model for the simulation of epileptic seizures

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A coupled ordinary differential equation lattice model for the CA3 region of the hippocampus (a common location of the epileptic focus) is developed. This model consists of a hexagonal lattice of nodes, each describing a subnetwork consisting of a group of prototypical excitatory pyramidal cells and a group of prototypical inhibitory interneurons connected via on/off excitatory and inhibitory synapses. The nodes communicate using simple rules to simulate the diffusion of extracellular potassium. Both the integration time over which a node's trajectory is integrated before the diffusional event is allowed to occur and the level of inhibition in each node were found to be important parameters. Shorter integration times lead to total synchronization of the lattice (similar to synchronous neural activity occurring during a seizure) whereas longer times cause more random spatiotemporal behavior. Moderately diminished levels of inhibition lead to simple nodal oscillatory behavior. It is postulated that both the lack of inhibition and an alteration in conduction time may be necessary for the development of a behaviorally manifest seizure. © 1999 American Institute of Physics. [S1054-1500(99)00403-6]

Complex partial epileptic seizures are thought to originate from an area of focal abnormality in the brain, usually in the hippocampal portion of the temporal lobe.¹ The abnormal firing patterns exhibited by this focal region spread throughout surrounding normal brain tissue using a poorly understood recruitment mechanism. Neither the precise origin of the abnormal firing activity in the focus nor the physico-chemical nature of the recruitment mechanism are understood. Surgical removal of the small group of cells in the focal region can often, but not always, eliminate seizures of this type. The reason the surgery fails in some cases is not understood and even in the case of "success," serious side effects and unintended injuries can result if too much tissue is removed or the location of the focus is not accurately determined. In this work we report on the creation and study of a mathematical model of the hippocampal region which exhibits firing activity similar to that observed clinically. By studying the dynamical properties of this model, particularly the features involved with spread of seizure-like activity, we have been able to identify a parameter we call t_{int} that may be abnormally altered in the healthy tissue surrounding the focus, allowing seizures to spread from the abnormal focus to the whole brain. When this parameter is small, communication between neighboring groups of otherwise healthy neurons is abnormally fast and seizures may result; when t_{int} is larger, communication is slowed down and seizures originating in the abnormal focus might not be able to spread. While much further study is needed, we speculate that the parameter t_{int}

might reflect changes in communication speeds through the glial network. Other possibilities for a variable communication speed include changes in extracellular ion concentrations that arise from variation in extracellular volume. All these possibilities suggest new, nonsurgical approaches to treatment of the disease.

I. INTRODUCTION

The incidence of epilepsy in developed nations is about 7 cases/1000 population² and the most common type in adults leads to seizures classed as partial.^{3,4} As many as 5–10% of patients with epilepsy will eventually become resistant to medication;⁵ however, many in this latter group can be helped by operative intervention to remove the focal generator site. We have found that this surgery can render 90% of properly-selected patients seizure-free.⁶ As noted above, it is not clearly understood why the surgery succeeds nor is it understood why the procedure fails in the 10% of intractable cases. Furthermore, it is likely that a portion of brain tissue much larger than the actual focus is (unnecessarily) removed, leading to impairment in a variety of functions. Therefore, it is of great clinical importance to understand more thoroughly the dynamics of seizure generation and spread.

In the current paper, we introduce a three-variable ordinary differential equation (ODE) model for a hippocampal subnetwork believed to be of importance in the generation of focal, or complex partial, seizures. The variables in this model correspond to membrane potentials for prototypical pyramidal cells and inhibitory interneurons in the CA3 region of the hippocampus, the most likely location of the focus. These prototypical cells are assumed to describe the

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average or mean behavior of a population of ~ 1000 or so neurons in a localized portion of the CA3 region. The averaging inherent in this definition is most apparent in the functional form assumed for the excitatory and inhibitory synaptic connections between the prototypical cells; the choice of smooth sigmoidal functions (as opposed to “on/off” step functions expected for single cells) implies an average over a population or group. Wilson and Cowan in 1972⁷ used a similar hyperbolic function for the same purpose in their model for reciprocally connected small groups of neurons.

The subnetwork ODE model, consisting as it does of interconnected excitatory and inhibitory elements, is similar to a long line of models studied by others^{7–12} and would, thus, be expected to exhibit the type of dynamics occurring in the CA3 region. We present a summary of its dynamical properties here (see Ref. 13 for further details) but focus on the spatiotemporal behavior of a coupled lattice model in which each cell (or node) of the lattice is described by the subnetwork ODE model. Coupling of the nodes in the lattice is done in a way that is more physically realistic than has been typical in model networks; here we assume coupling via potassium diffusion. Diffusion of extracellular potassium has been proposed as a possible mechanism of synchronization of neurons and the consequent aberrant neural recruitment during the propagation of a seizure.^{8,14–18}

Thus, the lattice which we present here models the spatiotemporal behavior of a macroscopic area of brain, possibly the entire CA3 region of the hippocampus. One of the questions which has motivated our study of this problem is the following: how large is the focal region? At present, most operations remove the entire anterior 4.5 to 6.0 cm of the affected temporal lobe, a quantity of tissue that is likely much larger than the size of the structurally abnormal focus. Surgery can lead to complications such as memory loss, speech deficit and visual impairment. Hence, it is of enormous practical interest to develop better methods for localizing and estimating the size of the focus in patients for whom surgery is planned. We believe that the lattice model proposed here will lead to insight into the size of the focus needed for the generation of seizure-like activity in the model.

The effect of disinhibition of the lattice on its dynamics was also of interest in our investigation. It has been proposed that a lack of inhibition mediated either by a reduction in the number of excitatory cells that synapse on the inhibitory cells^{19,20} or the reduction of the release of the inhibitory neurotransmitter γ -amino-butyric acid (GABA)²¹ leads to seizures. Other investigators have found inhibition in networks of neurons to be an important parameter controlling synchronization.^{22,23} Additionally, the speed with which communication occurs between populations of neurons should be an important factor in seizure propagation. It is expected that faster communication between neurons would facilitate synchronous neural activity. These two factors are investigated by varying certain parameters in the model, which will now be described.

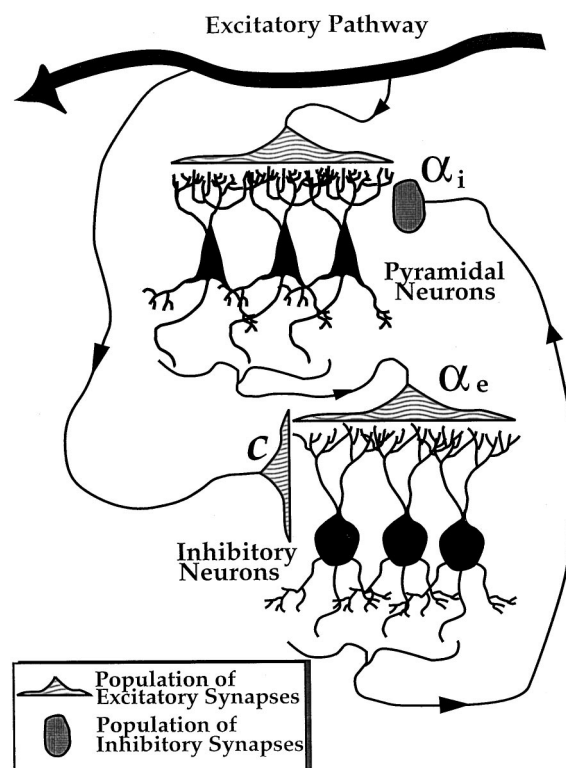


FIG. 1. A representative subnetwork showing interconnections between an excitatory pathway (the perforant path of the CA3 region of the hippocampus), a population of pyramidal cells and a population of inhibitory interneurons. The parameter c governs the strength of the current input to the inhibitory interneurons and the factors α_{exc} and α_{inh} govern the strength of the excitatory and inhibitory synaptic connections, respectively.

II. MATERIALS AND METHODS

The model we developed is shown schematically in Fig. 1. Two types of neurons are included in the subnetwork which forms the basis of the model: pyramidal cells with membrane potential V_i and inhibitory interneurons with membrane potential Z_i interconnected by synapses and both fed by current from the excitatory (i.e., perforant) pathway. The subnetwork corresponds to a small group of neurons located at one spatial coordinate “ i ” in the lattice. The dynamical behavior of this subnetwork is described by a system of three differential equations based on a two-variable reduction of the Hodgkin–Huxley model^{24,25} initially proposed by Morris and Lecar^{26,27} as a model for barnacle muscle fiber, but of general utility in modeling the pyramidal cells in a subnetwork of the type shown in Fig. 1. To model the behavior of the entire subnetwork (i.e., pyramidal cells plus inhibitory interneurons), we have added an equation to the Morris–Lecar model to simulate the effect of a population of inhibitory interneurons synapsing on the pyramidal cells. Thus, the model consists of an interconnected population of excitatory and inhibitory elements, both of which are fed by an external current. The latter feature, while physiologically realistic, has apparently not been included in prior similar models but turns out to be useful for studying the effects of disinhibition on the model dynamics. The system of equations for our proposed subnetwork model is

$$\frac{dV_i}{dt} = -g_{Ca}m_\infty(V_i - 1) - g_KW_i(V_i - V_i^K) - g_L(V_i - V^L) + I - \alpha_{inh}Z_i, \quad (1)$$

$$\frac{dW_i}{dt} = \frac{\phi(w_\infty - W_i)}{\tau_w}, \quad (2)$$

$$\frac{dZ_i}{dt} = b[cI + \alpha_{exc}V_i]. \quad (3)$$

The Morris–Lecar model consists of Eqs. (1) and (2) with the last term of Eq. (1) (i.e., $\alpha_{inh}Z_i$) missing. The first and third variables, V_i and Z_i are, again, the membrane potentials of the pyramidal and inhibitory cells, respectively, while the second variable, W_i , is a relaxation factor which is essentially the fraction of open potassium channels in the population of pyramidal cells; all three variables apply to node i in the lattice. The parameters g_{Ca} , g_K and g_L are the total conductances for the populations of Ca, K and leakage channels, respectively. V_i^K is the Nernst potential for potassium in node i ; this parameter will be used in coupling subnetwork populations together into a lattice. V^L is a leak potential, τ_w is a voltage dependent time constant for W_i , I is the applied current, and ϕ and b are temperature scaling factors. The parameter c in Eq. (3) differentially modifies the current input to the inhibitory interneuron.

Other important variables in Eqs. (1)–(3) are the following functions of V_i and Z_i :

$$m_\infty = 0.5 \left[1 + \tanh\left(\frac{V_i - V_1}{V_2}\right) \right], \quad (4)$$

$$w_\infty = 0.5 \left[1 + \tanh\left(\frac{V_i - V_3}{V_4}\right) \right], \quad (5)$$

$$\alpha_{exc} = a_{exc} \left[1 + \tanh\left(\frac{V_i - V_5}{V_6}\right) \right], \quad (6)$$

$$\alpha_{inh} = a_{inh} \left[1 + \tanh\left(\frac{Z_i - V_7}{V_6}\right) \right], \quad (7)$$

$$\tau_w = \left[\cosh\left(\frac{V_i - V_3}{2V_4}\right) \right]^{-1}. \quad (8)$$

The functions w_∞ and m_∞ which appeared in the original Morris–Lecar model are nondimensionalized expressions that describe the voltage-regulated Ca^{2+} channels in the cell membrane, each of which is either open or closed at any given moment. The hyperbolic tangent function provides a simple way to model the collective behavior of a large number of channels while retaining the open/closed (or on/off) character of the individual channels. In Hodgkin–Huxley type equations m_∞ is a type 1 variable assumed to have a more or less instantaneous onset (i.e., a signum function), whereas w_∞ is type 3 with a much more gradual upstroke. We assume that a population of neurons, each of which is behaving in this fashion, will tend to show a decrease in the slope of both variables through spatiotemporal dispersion and Eqs. (4) and (5) reflect this assumption. We have, however, used appropriate parameters so as to maintain the rela-

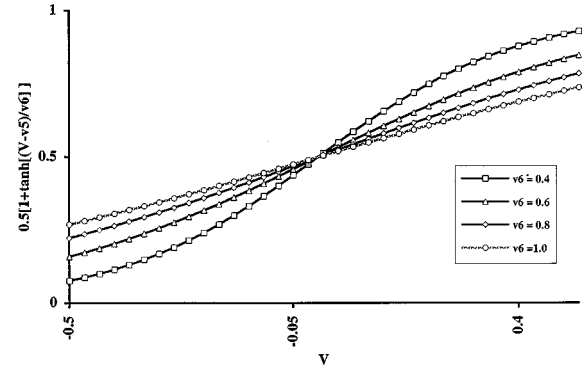


FIG. 2. The excitatory synaptic connection α_{exc} given by Eq. (6) for $V_6 = 0.4, 0.6, 0.8$ and 1.0 . Here, $V_5 = 0.0$.

tive relationship between the slopes with that of m_∞ being much greater than for w_∞ . We take a similar approach in modeling the populations of excitatory and inhibitory synaptic connections between pyramidal cells and their interneurons in the population of cells corresponding to the subnetwork model. Thus, the functions α_{exc} and α_{inh} are also taken to have a hyperbolic tangent form parametrized by the dimensionless synaptic strengths a_{exc} and a_{inh} , respectively, a steepness parameter (V_6 for both α_{exc} and α_{inh}) and threshold parameters (V_5 for α_{exc} and V_7 for α_{inh}). As illustrated in Fig. 2, as the average or mean pyramidal cell potential V_i in the subnetwork population at node i increases, the fraction of excitatory synapses that are “on” or “open” also increases. The smaller the value of V_6 , the more step-like this response is. Hence, larger V_6 values correspond to a more heterogeneous population of neurons, each of which is interconnected via on/off excitatory and inhibitory synapses. The inhibitory synapse α_{inh} has the same functional properties as α_{exc} but the negative sign preceding it in Eq. (2) imparts an inhibitory effect from the interneuron, with membrane potential Z_i , to the pyramidal cell. As will be shown below, a great deal of insight into lattice dynamics in this model can be achieved by systematically varying the parameters in α_{exc} and α_{inh} while carrying out the simulations.

The effect of including a time delay in the model was studied and found to have no effect on the qualitative features of the results. However, the lattice simulation method (introduced below) includes an integration time, t_{int} , over which the subnetwork model is allowed to run before spatial coupling occurs; its effect may be similar to a time delay.

The dynamical properties of the subnetwork for a single spatial coordinate “ i ” were systematically studied; the results are summarized in the following section. Because the subnetwork dynamics seemed to be similar to what might occur in the hippocampus, we created a spatially extended model in order to study the ability of these subnetworks to become synchronized, as in a seizure. The spatially extended model is created by coupling together a number of nodes (subnetworks), each of which corresponds to a group of neurons governed by Eqs. (1)–(3). The nodes are coupled in a hexagonal lattice with periodic boundary conditions as shown in Fig. 3 via rules used to simulate the diffusion of potassium ions. A hexagonal lattice was used because it has

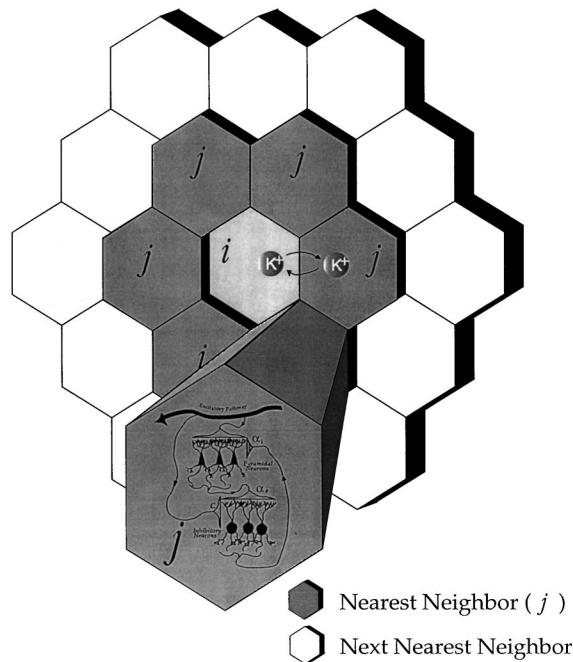


FIG. 3. Hexagonal lattice of cells. Communication between cells is mediated by potassium ion diffusion between nearest neighbor cells (cells “*j*”). Thus, the parameter V_i^K for cell “*i*” is a function of the extracellular potassium concentration in the six nearest neighbor cells (cell “*j*”). The blown-up cell emphasizes that the dynamics at each node is governed by the same type of subnetwork, i.e., the one involving interactions between the two populations of neurons shown in Fig. 1.

six equivalent nearest neighbors whereas a square lattice has eight nearest neighbors of two different types: four connecting at the sides and four connecting at the corners. Preliminary studies with a square lattice indicated that the spatiotemporal behavior was strongly influenced by the lattice symmetry; hence, a hexagonal lattice was used in this study.

The choice of mechanism for coupling the nodes is not a trivial problem. At least three mechanisms seem biophysically reasonable: (a) Diffusive; (b) delay line (axonal/synaptic or glial); and (c) extracellular current flow. Each of these processes is at some level a function of the intra- and extracellular $[K^+]$ which in turn is reflected in the potassium equilibrium potential (V_i^K). Thus, we have chosen to use this variable as the coupling element between nodes.

It will be noted that coupling based on the three mechanisms above varies over several orders of magnitude, from 2–5 mm/min for diffusive to 10–20 m/s for delay line transmission. Thus, a parameter, t_{int} , was chosen which can be varied to look at alterations in transmission time (coupling). It functions, then, essentially as a delay term. Furthermore, since we are interested primarily in the dynamical properties of the system, we make no claim that K^+ diffusion is the actual mechanism of communication from one subnetwork to another, but simply that the K^+ equilibrium potential is likely to be a marker for the process, whatever it is. In this respect, our approach is similar to that of Kaneko.²⁸

Thus, each node is governed by the system of equations Eqs. (1)–(3) representing the subnetwork in Fig. 1 and is integrated to determine the time dependent membrane potential of the pyramidal neurons in that node (V_i). These inte-

grations are performed using a 4th order Adams–Bashforth predictor method with a 4th order Gear corrector programmed in C. A fixed time step of 0.05 time units was used. Convergence and the absence of numerical instabilities was ensured by checking the robustness of the results using different integrators, different lattice sizes, etc. No evidence of numerical problems was detected. Transients are allowed to decay for a length of time, $t_{\text{int}}/6$; additionally, t_{int} determines how long the node’s dynamics will remain autonomous, without communication with other nearest neighbor nodes. The quantity t_{int} is, therefore, a measure of the speed of communication between nodes (mediated here by potassium ion diffusion). Exploration of the effects of altering t_{int} yields important results discussed below. At the beginning of each simulation, two nodes are spiked with excess extracellular potassium and spatially random initial conditions are used. Since neuronal activity leads to the extrusion of potassium into the extracellular space, a time averaged potential ($\bar{V}_j$) is used to determine how much additional potassium is contributed to the extracellular environment by the pyramidal cells at each node in the hexagonal lattice. The parameter, V_i^K , the equilibrium potential of potassium for node “*i*,” is taken to be a function of the average extracellular potassium (or, equivalently, the average membrane potential $\bar{V}_j$) of the six nearest neighbors “*j*.” The value of V_i^K is changed after each interval of time using the following equation:

$$V_i^K = \left(\sum_{j=1}^6 \frac{\bar{V}_j}{6} \right) - \frac{1}{2}, \quad (9)$$

where

$$\bar{V}_j = \frac{1}{T} \int_{t_{\text{int}}/6}^{t_{\text{int}}} V_j(t) dt, \quad (10)$$

and $T = t_{\text{int}} - t_{\text{int}}/6$ is the interval of time used in computing the averages. Thus, the spatially averaged pyramidal cell membrane potential for each of the nearest neighbor subnetworks, V_j , was obtained by solving Eqs. (1)–(3) for node “*j*.” The time averaged potential for the delay interval, t_{int} , was then calculated using Eq. (10). The lower integration limit, $t_{\text{int}}/6$, was selected to allow time for transients to decay. Finally, the potential in node “*i*” was computed from Eq. (9). Equation (9) was obtained by, first, restricting the parameter V_i^K to values that placed a single node “*i*” of the system in a temporal bursting mode. The average potential $\bar{V}_j$ of the six neighboring nodes was then found and it was noted that a straight line equation related V_i^K to $\bar{V}_j$ with a slope of 1 and an intercept of 1/2. Equation (9), then, yields the updated potassium ion Nernst potential V_i^K which is subsequently used in Eqs. (1)–(3) for each node over the next interval of time.

This system is consequently a hybrid of a continuous differential equation model and a discrete cellular automaton, which uses decision rules for coupling nodes in a lattice similar to the rule employed here for the diffusion of potassium. As noted before, diffusion of extracellular potassium has been implicated in the synchronization of neurons and

TABLE I.

Parameter	Description	Value
V_1	threshold value for m_∞	-0.01
V_2	steepness parameter for m_∞	0.15
V_3	threshold value for w_∞	0.0
V_4	steepness parameter for w_∞	0.30
V_5	threshold value for α_{exc}	0.0
V_6	steepness parameter for α_{exc} and α_{inh}	varied (see figures)
V_7	threshold value for α_{inh}	0.0
g_{Ca}	conductance of population of Ca^{2+} channels	1.1
g_{K}	conductance of population of K^+ channels	2.0
g_{L}	conductance of population of leak channels	0.5
V_i^{K}	equilibrium potential of K^+	-0.7
V_i^{L}	equilibrium potential of leak channels	-0.5
τ_w	time constant for W_i	1.0
I	applied current	0.30
b	time constant scaling factor	varied (see figures)
c	strength of feedforward inhibition	varied (see figures)
a_{exc}	strength of excitatory synapse	varied (see figures)
a_{inh}	strength of inhibitory synapse	varied (see figures)
ϕ	temperature scaling factor	0.7

may, therefore, play a role in the consequent aberrant neural recruitment during the propagation of a seizure.^{8,14–18}

III. RESULTS

The dynamical properties of the subnetwork for a single spatial coordinate “ i ” were systematically studied; the results are reported in detail elsewhere (see Ref. 13) but will be summarized here. The system of equations was numerically solved using the function NDSolve in MATHEMATICA (Wolfram Research) on an Indigo II Silicon Graphics workstation and later confirmed using the in-house Adams–Bashforth routine described earlier. Table I summarizes the values of the parameters used in the simulations; other specific values are given in the figure captions. We found that the model displays a rich variety of different dynamical behaviors. Rinzel and Ermentrout have previously shown²⁷ that the Morris–Lecar model can reproduce single action potentials as well as sustained limit cycle oscillations for different choices of parameter values. In simulations with the modified model for the subnetwork, Eqs. (1)–(3), we find simple limit cycles as well as complex periodic oscillations and aperiodic behavior. Figure 4 shows the time series for the three variables for a typical complex periodic state. The three-dimensional phase portrait is also shown [Fig. 4(d)] and it appears that the complex periodic state corresponds to a phase-locked mixed-mode state on a torus attractor. As the parameter c (the current strength of the perforant path input to the inhibitory cell) is varied we see a large variety of mixed mode states interspersed with regions of apparent chaotic behavior. Figure 5 summarizes the dependence of the dynamics of this model on the parameter c as a bifurcation diagram in which the maxima in the V_i time series are graphed for various values of the parameter c . In general, the dynamics are less complex for smaller values of “ c ,” i.e., when the current to the inhibitory cells is relatively low compared to the current input to the pyramidal cells. Hence, a low degree of inhibition results in a system that is more

likely to be periodic, suggesting the type of spatiotemporal coherence that could exist with a seizure.²⁹ When inhibition is completely absent, however, the dynamics of the system goes to a fixed point corresponding to a state of total depolarization of the subnetwork. A close inspection of the transition regions between mixed-mode periodic states and chaos indicates that the mixed-mode states undergo a period-doubling sequence as the underlying torus attractor breaks up into a fractal object. Similar behavior is seen in a series of systematic investigations when other parameters are varied, such as b which governs the time scale of the inhibitory interneuron and V_6 which governs the steepness of the on/off switch in the population of synapses. See Ref. 13 for further details.

A. Effect of averaging time

In order to model lattice connections in terms of potassium ion diffusion, a decision must be made about the time scale for diffusion. Diffusion of potassium from one cell into a neighboring cell is not instantaneous, but occurs on a characteristic time scale which is related to the diffusion coefficient. As noted above the time scale for communication of one subnetwork with another will vary markedly depending upon the exact mechanism operative at any given time. We have carried out a study in which we vary the integration time, t_{int} , i.e., the time over which a given node’s trajectory is integrated before the diffusional event is allowed to occur. We find that synchronization of the lattice occurs when this integration time falls below a certain critical value, $t_{\text{int}} = 13.5$ time units (13.5’). Note that time units are strictly arbitrary. The use of the quantity ‘ does not imply minutes. For values of t_{int} greater than or equal to this minimum, the lattice remains unsynchronized as indicated by subtracting the average pyramidal potentials, $\bar{V}_i$, for two typical but arbitrary nodes in the lattice (nodes 149 and 814 shown in Fig. 6). Figures 7 and 8 show the effect of the integration time on the average pyramidal potential for node 814 and the

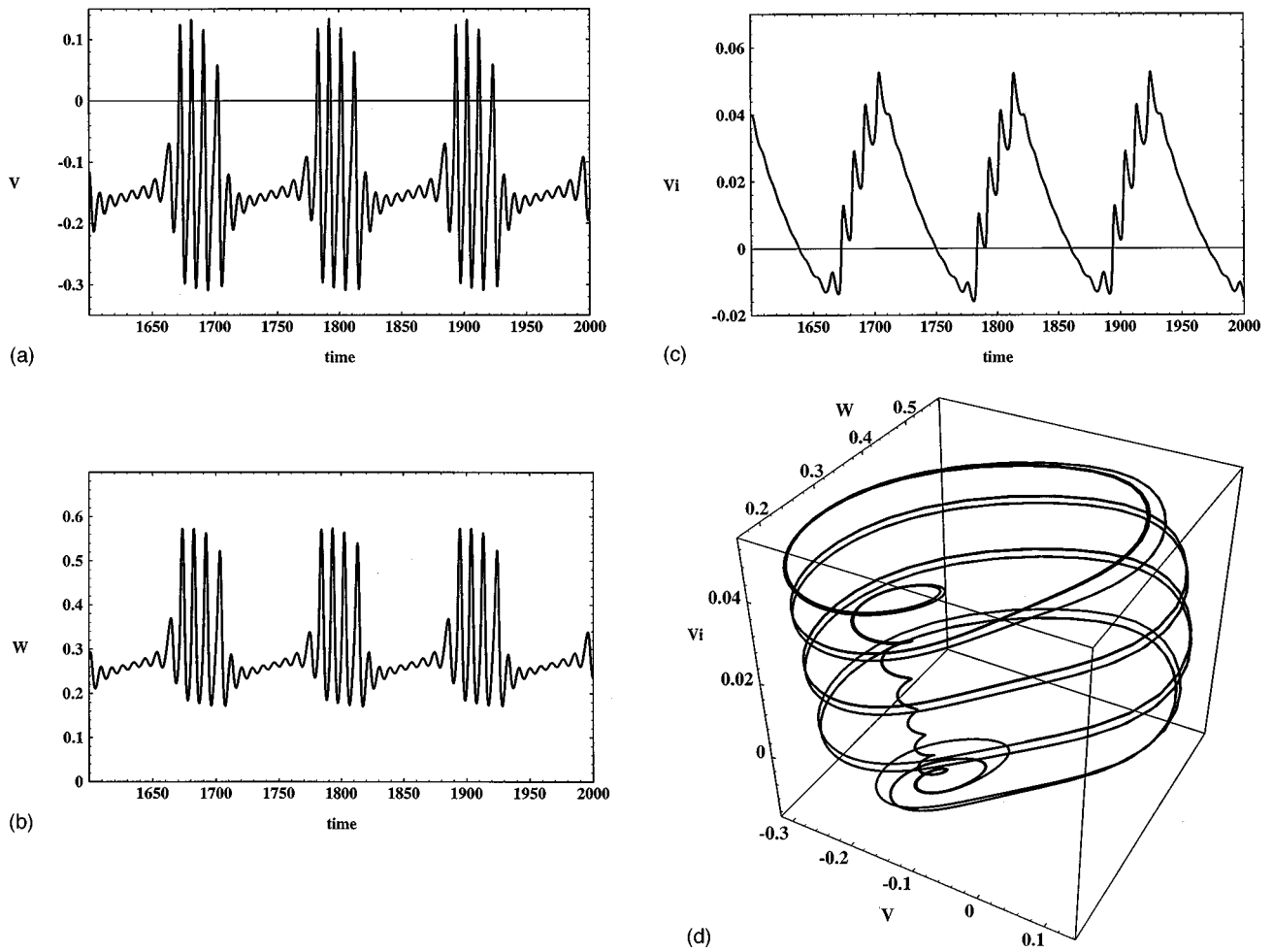


FIG. 4. (a) V_i ; (b) W_i ; and (c) Z_i vs t . (d) Phase portrait in the V_i , W_i , Z_i space. Parameter values are $V_1 = -0.01$, $V_2 = 0.15$, $V_3 = 0.0$, $V_4 = 0.30$, $V_5 = 0.0$, $V_6 = 0.6$, $V_7 = 0.0$, $g_{Ca} = 1.1$, $g_K = 2.0$, $g_L = 0.5$, $V_1^K = \text{varied}$; $I = 0.30$, $b = 0.1$, $f = 0.7$, $a_{exc} = 1.0$, $a_{inh} = 1.0$ and $c = 0.165$.

difference in average potential between node 814 and node 149 (i.e., $\bar{V}_{814} - \bar{V}_{149}$). With an integration time of 12 arbitrary time units ($12'$), all of the nodes become synchronized in a spatially uniform periodic limit cycle [Fig. 7(a)]. As the

value of t_{int} is increased to $13.5'$, a 1:1 mixed-mode limit cycle appears for the time series of node 814 and the potential difference plot shows that the two nodes are slightly out of phase [Fig. 7(b)]. As the value of t_{int} is increased further, more complicated mixed-mode states appear, although they are all still periodic [Figs. 7(c) and 7(d)]. However, at $t_{int} = 42'$ the time series for node 814 has become slightly aperiodic.

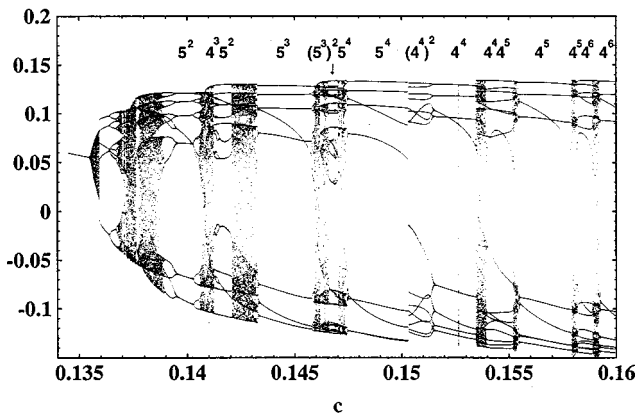


FIG. 5. A bifurcation diagram showing the global behavior of the maxima of V_i at many values of c . Mixed-mode states are labeled above the applicable parameter interval as L^s , where L is the number of large amplitude oscillations and s is the number of small amplitude oscillations per cycle. Parameters are given in Fig. 4 with c variable.

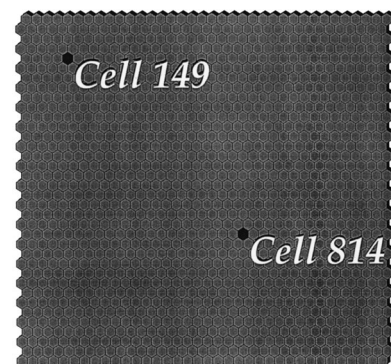


FIG. 6. A 36×36 hexagonal lattice showing the location of nodes 149 and 814 whose pyramidal potential values are subtracted to indicate the degree of lattice synchronization in Figs. 7, 8 and 11.

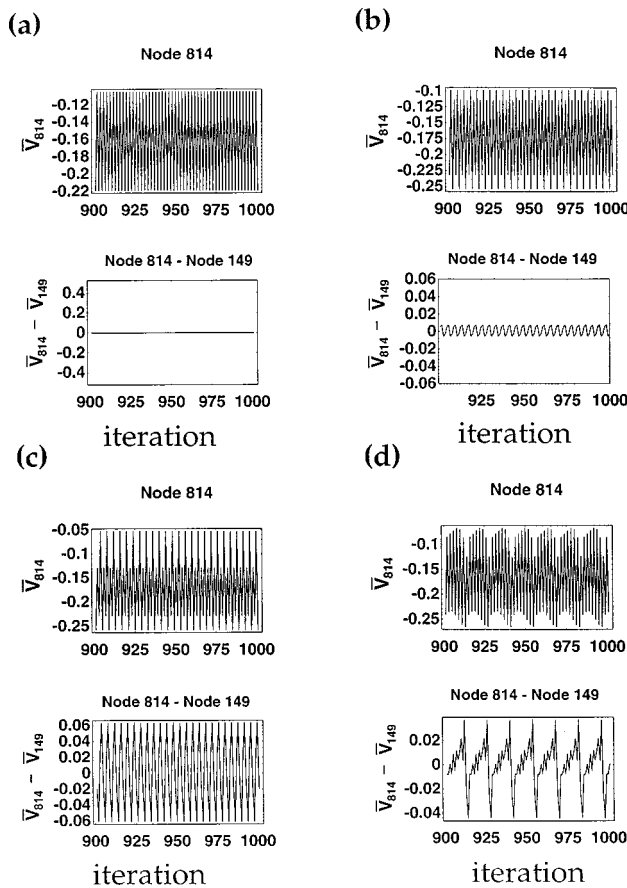


FIG. 7. The average pyramidal potential as a function of time (iterations 900–1000) and the average potential difference between node 814 and node 149 as a function of time for (a) $t_{int} = 12'$; (b) $t_{int} = 13.5'$; (c) $t_{int} = 15'$; and (d) $t_{int} = 18'$. Other parameter values used to obtain (a)–(d) are given in Fig. 4.

riodic and the lattice is now unsynchronized [see Fig. 8(a)]. However, note that the two nodes are nearly synchronized, as can be seen by the small amplitude of the values. At $t_{int} = 42.6$ [see Fig. 8(b)] the time series is fully aperiodic and the difference between node 149 and node 814 has a much greater amplitude. However, the difference between node 554 and node 566 is not as large [see Fig. 8(c)], indicating that the degree of synchronization depends on the spatial location of the nodes on the lattice. Additionally, the time series of node 554 appears much more periodic than that of node 814 indicating that for this value of t_{int} , the lattice dynamics is one probably described as a periodic state with a great degree of spatial complexity. At $t_{int} = 48'$ full aperiodicity, both spatially and temporally, has appeared [Fig. 8(d)].

In order to better understand the complex behavior seen near the value of $t_{int} = 42.6$, we need to take a look at the entire lattice. Figure 9 shows the average pyramidal potential using a gray scale for each of the 1296 nodes of the lattice at consecutive iterations for various values. In Fig. 9(a), the lattice is fully synchronized at $t_{int} = 12'$. As the value of t_{int} increases [Fig. 9(b)–9(f)] standing wave patterns, some more spatially complex than others, appear in sequence. At $t_{int} = 48'$ [Fig. 9(g)], a spatially aperiodic state is found indicating that the lattice is fully desynchronized as was suggested by Fig. 8(d).

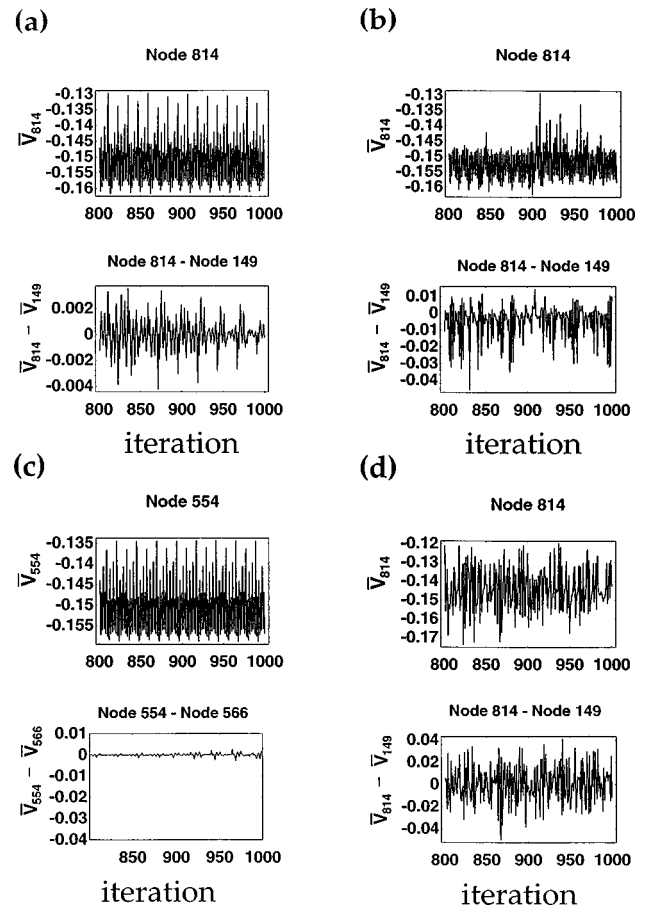


FIG. 8. The average pyramidal potential as a function of time (iterations 800–1000) and the average potential difference between node 814 and node 149 for (a) $t_{int} = 42'$; (b) $t_{int} = 42.6'$; (c) $t_{int} = 42.6'$ (average potential difference between node 554 and node 566); and (d) $t_{int} = 48'$. Parameter values are as given in Fig. 4.

Figure 10 shows a three-dimensional graph of the average pyramidal potential over the 36×36 node lattice. Once again, increasing t_{int} from $42'$ [Fig. 10(a)] to $42.6'$ [Fig. 10(b)] and, finally, to $48'$ [Fig. 10(c)] shows how the irregularity of the spatial pattern increases as this parameter is increased and suggests that a bifurcation to full spatiotemporal chaos occurs near $t_{int} = 42.6$.

B. Effect of inhibition

We also studied the effect of the ratio of inhibition to excitation on the seizure mimicking capacity of our lattice model. In studies of the dynamics of the subnetwork reported in detail elsewhere (see Ref. 13), we modeled the ratio of inhibition to excitation by varying the parameter “ c ” which governs the ratio of the current input to the inhibitory interneuron and pyramidal cell, respectively, and found that lower relative levels of input to the inhibitory cell tended to make the dynamics more ordered. This result is consistent with the experimental and clinical observations that reduced levels of inhibition (through either smaller numbers of interneurons or decreased inhibitory synaptic connections in the CA3 region of the hippocampus) are associated with the occurrence of seizures.^{19–21}

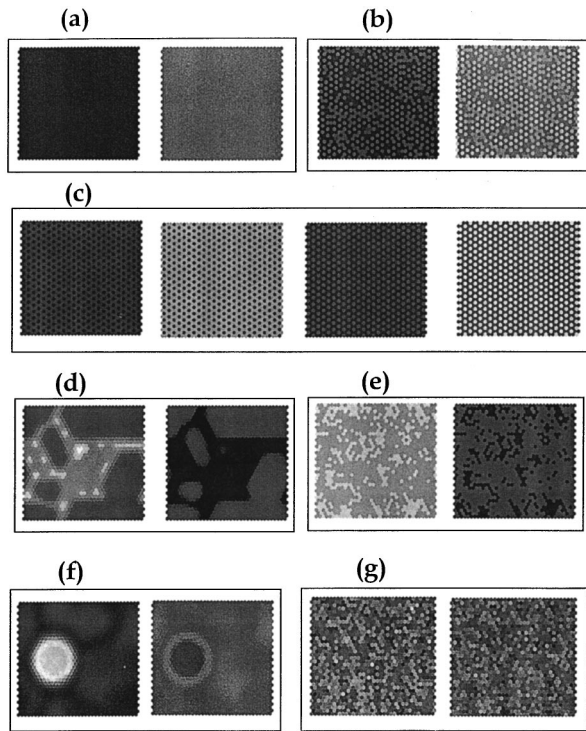


FIG. 9. Lattice patterns of sample consecutive iterations for (a) $t_{\text{int}} = 12'$; (b) $t_{\text{int}} = 13.5'$; (c) $t_{\text{int}} = 15'$; (d) $t_{\text{int}} = 18'$; (e) $t_{\text{int}} = 24'$; (f) $t_{\text{int}} = 42'$; and (g) $t_{\text{int}} = 48'$. In order to best resolve the spatial patterns, the gray scale corresponds to different values of $\bar{V}_i$ for each hexagonal lattice plot. Parameter values are as given in Fig. 4.

Here we have chosen to consider the synchronization capabilities of the lattice by varying, again, the ratio of inhibition to excitation. Now, however, we take a more direct approach by varying the parameters a_{inh} and a_{exc} which appear in the synaptic connections α_{inh} and α_{exc} [Eqs. (6) and (7)]. We find a strong dependence of the dynamics of the lattice on these parameters, as shown in Fig. 11. Both a_{inh} and a_{exc} were set equal and varied from $a_{\text{inh}} = a_{\text{exc}} = 0.7$ (reduced synaptic strength or reduced feedback inhibition) to $a_{\text{inh}} = a_{\text{exc}} = 1.3$ (increased synaptic strength or increased feedback inhibition). The potential differences between node 814 and node 149 and the spatial pattern of potential for the entire lattice are shown in Fig. 11. At $a_{\text{inh}} = a_{\text{exc}} = 0.7$ [Fig. 11(a)] the lattice is totally “synchronized” in that every node has the same potential, but the entire lattice is at steady state and not firing at all. At $a_{\text{inh}} = a_{\text{exc}} = 1.0$ [Fig. 11(b)] the difference time plot displays a noisy signal with an underlying high amplitude, slower frequency component. At $a_{\text{inh}} = a_{\text{exc}} = 1.3$ [Fig. 11(c)] the difference time plot shows an aperiodic region followed by a more periodic one. The corresponding lattice graphs show that although the time series of the potential differences between two cells can be very different [figs. 11(b) and 11(c)], the spatial patterns of the corresponding lattice can show a similar disorder and are difficult to distinguish. The main conclusion is that increasing the degree of inhibition by increasing a_{inh} and a_{exc} from 0.7 to 1.0 causes the lattice to become totally unsynchronized, but the precise value at which this occurs cannot yet be determined.

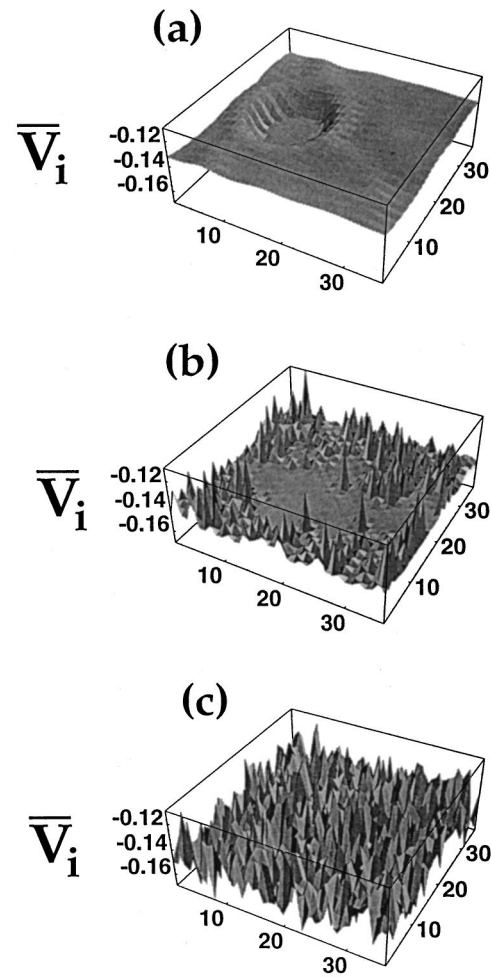


FIG. 10. Three-dimensional plots of the average pyramidal potential for the lattice of 1296 cells for (a) $t_{\text{int}} = 42'$; (b) $t_{\text{int}} = 42.6'$; and (c) $t_{\text{int}} = 48'$. Parameter values are as given in Fig. 4.

IV. DISCUSSION

The amount of time that each of the nodes in the lattice evolves autonomously, t_{int} , before a given node undergoes a discrete change of the parameter V_i^K is an important factor in this study. Qualitatively, the lattice appears to undergo two bifurcations as this control variable is changed. The first bifurcation at a t_{int} value between $12'$ and $13.5'$ takes the lattice from synchronized network activity to periodic standing wave patterns. The standing waves correspond to a roughly alternating sequence of periodic and aperiodic states; although we have not characterized the temporal nature of the lattice in detail, this is qualitatively in accord with the time dependent behavior of the equations for each separate node. The second bifurcation at a t_{int} value near $42'$ occurs as the lattice moves from these spatially ordered patterns to a spatially disordered and temporally aperiodic state. This bifurcation sequence of the spatial patterns is similar to some of the universal sequences found in the dynamics of coupled map lattices³⁰ which is not surprising since these abstract models capture the excitability properties being modeled here in a more physically realistic way.

The underlying presumption in this study is that the brain is a very complex nonlinear dynamical system that can

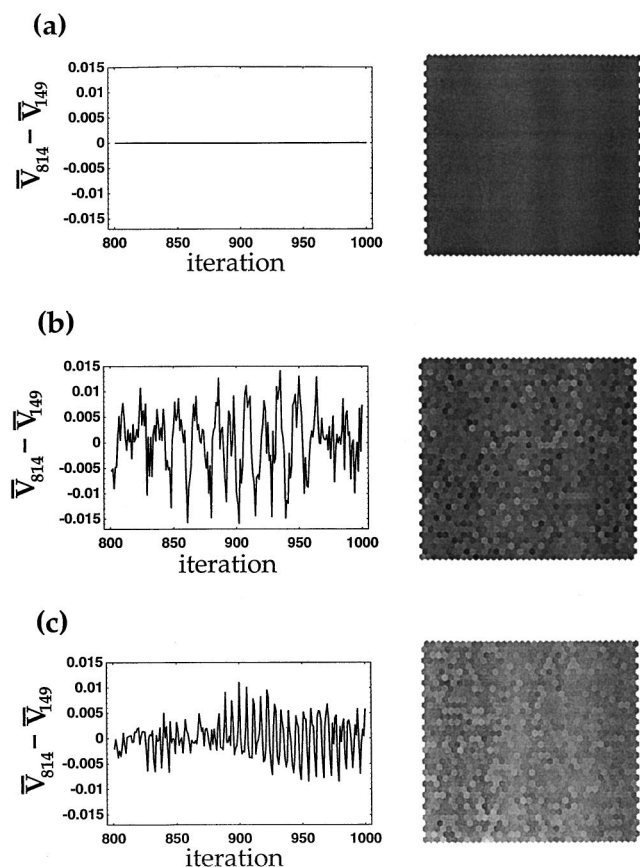


FIG. 11. Average pyramidal potential difference of node 814 and node 149 as a function of time (iteration 800–1000) and lattice graphs showing the average pyramidal potential of each cell, for (a) $a_{inh} = a_{exc} = 0.7$; (b) $a_{inh} = a_{exc} = 1.0$; and (c) $a_{inh} = a_{exc} = 1.3$. Parameter values are as given in Fig. 4 except $t_{int} = 120'$.

display behavior which is periodic or chaotic of varying dimensionality.^{7,31,32} It has been suggested that chaotic behavior in the brain is healthy whereas lower-dimensional chaos or periodicity is an indicator of disease.³³ In the case of epilepsy, the current study shows that if populations of neurons are communicating quickly (a small value of t_{int}) the nodes of the lattice synchronize in a periodic limit cycle, much like a seizure where many neurons are synchronized. As the integration time is increased, the lattice shows more complicated behavior that should correspond to healthier neural tissue. Freeman and Skarda³⁴ and subsequently Kelso and Fuchs³⁵ have suggested that chaos is desirable in brain activity because a chaotic state corresponds to an infinite number of unstable periodic orbits which would then be quickly available for neural computation.

Recent experimental observations provide several possibilities for physiological interpretation of the parameter t_{int} . This parameter in our model is correlated with increased speed or ease of communication between populations of neurons. Heinemann *et al.*^{36,37} have demonstrated changes in extracellular K^+ and Ca^{2+} in animal models of chronic epilepsy and Lux *et al.*³⁸ have shown alterations in the size of the extracellular space in epileptic tissue. These events presumably alter electrical resistance and field properties of the system resulting in increased conductance which facilitates

the entrainment of adjacent areas by the abnormal focus. Hence, t_{int} could be interpreted as a change in extracellular ion concentration (either K^+ or Ca^{2+}). Dudek,³⁹ in fact, has proposed that intervention to block the activity-induced cellular swelling which contributes to the above changes may be an effective method of treating seizures. In addition, Crill^{40,41} has suggested that dynamic alteration of the dendritic length constant (λ) may occur which could provide a mechanism for directly modifying the speed of synaptic communication again providing a mechanism for the abnormal region to recruit its neighbors. In the current study we assumed potassium diffusion as the basis of spatial coupling, i.e., the mechanism by which synchronization occurs. Other mechanisms are possible, including coupling via synapse or through the surrounding glial network. These would imply a different biophysical interpretation of t_{int} and will be the subject of future investigations.

Focal epilepsy is a truly devastating illness for those so afflicted. Thus, it is important to speculate on the biophysical relevance of the results presented in this study. We have shown that by varying the amount of inhibition in a network with recurrent connections, the system can be moved from chaotic to periodic to, finally, steady state behavior. These results correlate well with experimental data from other investigators that demonstrate electrophysiological seizures with reduced inhibition.^{42–45} We also have demonstrated that it is the change in communication time from one region to the next, t_{int} , rather than a global alteration in inhibition which leads to spatiotemporally coherent behavior in the model. We would postulate, then, that in patients with focal epilepsy, altered inhibitory influence in the focus leading to pathological periodic behavior is a necessary *but not sufficient* condition for the production of a behavioral seizure in the patient. A temporary alteration in the conduction time from one region to the next is also necessary in order for the abnormal focus to produce the widespread spatiotemporal coherence in (partially) normal neurons that is characteristic of a clinical seizure. Such an alteration in t_{int} could result biologically from a change in the mechanism of spatial coupling, as well as from a change in the physical parameters for any specific mechanism. Obviously, it is important that these speculations be biologically testable. We have plans to look at cortical impedance and interstitial potassium and calcium concentrations in both a kindled rat model of focal epilepsy as well as in hippocampal slices removed from humans at epilepsy surgery. In fact, with ion selective electrodes, these investigations may even be extendable to intraoperative evaluations at the time of surgery itself. Furthermore, it would be interesting to look at paired neuronal recordings either in slices or in the rat model to assess the dynamics of delay line transmission. The results from this model and these experimental data then should help explain why patients with presumably structurally and functionally static focal abnormalities do not seizure constantly as well as why some patients who have a demonstrated focal abnormality on an electroencephalogram never have seizures.

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